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How Does Farmers' Digital Literacy Affect Green Grain Production?

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Abstract

Grain production is crucial for national security and stability. Studying the impact of digital literacy on green production by grain farmers is of great significance for ensuring food security and achieving green agricultural development. This article utilizes data from the 2020 China Rural Revitalization Survey (CRRS), selecting a sample of 1811 farming households engaged in grain cultivation. Employing methods such as the ordered Probit model and mediating effect model, it analyzes the impact of digital literacy on green grain production from the perspectives of transformation drivers and pathways. The results show, first, that digital literacy significantly promotes farmers' green production behaviors, and the findings remain valid after multiple robustness tests. Second, a mechanism analysis reveals that digital literacy drives farmers' green production by reconstructing their benefit cognition and green cognition and promoting the application of green mechanization technologies. Third, a heterogeneity analysis indicates that the larger the farmers' operation scale and the stronger their economic capacity, the more significant the promoting effect of digital literacy on their green production. Accordingly, it is necessary to accelerate the improvement of farmers' digital literacy, reduce green production costs, popularize green mechanization technologies, and promote the green transformation of grain production.

Keywords: digital literacy; green grain production; cognitive reconstruction; green mechanization technology

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1. Introduction

Food security is a top priority for the country. In recent years, China's grain production has achieved historic achievements, with the total output exceeding 0.7 trillion kg in 2024, remaining stable above 0.65 trillion kg for nine consecutive years. However, China's grain production has long relied on chemicals, such as fertilizers, pesticides, and agricultural films [1]. The utilization rates of fertilizers and pesticides for the three major grain crops are only 40.2% and 40.6%, respectively. Excessive input of chemicals not only exacerbates agricultural non-point source pollution but also damages the soil structure, reduces land quality, and affects the sustainability of grain production. In 2015, the former Ministry of Agriculture issued the Action Plan for Zero Growth in Fertilizer Use by 2020 and the Action Plan for Zero Growth in Pesticide Use by 2020. By 2022, it continued to issue the Action Plan for Fertilizer Reduction by 2025 and the Action Plan for Chemical Pesticide Reduction by 2025 to address agricultural pollution and protect the environment. Although this has, to some extent, inhibited the growth momentum of fertilizer and pesticide input

in China's grain production, the overall use intensity is still high [2], and the path of green production transformation in grain remains arduous. At the global level, the State of Food and Agriculture 2024 released by the Food and Agriculture Organization of the United Nations (FAO) points out that environmental impacts caused by unsustainable farming practices have significantly increased the hidden costs of food production, particularly in terms of greenhouse gas emissions, nitrogen runoff losses, changes in land use patterns, water pollution, etc. Therefore, it is urgent for food production to develop towards green and low-carbon directions. Under the basic pattern of small-scale peasant management, farmers are important behavioral subjects of agricultural production and management decisions [3]. Their green production behavior in the process of grain production is related to national food security and is also the micro-foundation of China's agricultural green development. Since 2019, China has vigorously promoted the construction of digital villages. The scale of rural Internet penetration has continued to expand: the total number of broadband users in rural China reached 192 million, the internet penetration rate in rural areas was 66.5%, and the size of the rural netizen population was 326 million, and farmers' digital literacy has been continuously improved. As an important component of farmers' human capital in the digital era, studying whether the improvement of digital literacy can promote green grain production is of great significance for ensuring food security and promoting the green development of agriculture.

The academic community has carried out rich research on farmers' green production behavior, and its influencing factors mainly include farmers' individual and family characteristics, management methods, cognitive characteristics, and other aspects. Farmers' age [4,5], gender [6], education level [7], planting area [8], family labor force size [9], etc., will affect farmers' green production behavior. In terms of management methods, promoting land transfer [10], increasing the scale of operation [11], encouraging farmers to join cooperatives [12] or participate in e-commerce [13], relying on land trusteeship [14] and socialized services [15,16], increasing agricultural technology training [17], etc., can promote farmers' green production. In terms of cognitive characteristics, farmers' values [18], risk perception [19,20], environmental awareness, social regulations [21], etc., also have a certain impact on green production.

In terms of food production, specifically, apart from the aforementioned influencing factors, Bidogeza et al. found that adopting alternative agricultural technologies in Rwanda can ensure food security and increase farmers' incomes [22]. Nakano et al. discovered significant potential for a green rice revolution in favorable rainfed rice-growing areas of sub-Saharan Africa through training on modern input use and improved agronomic practices [23]. Keyi et al. found that agricultural productive services can alleviate the constraints of agricultural production factors for smallholder farmers, reduce production costs, and improve the efficiency of green food production [24]. Meanwhile, with the vigorous development of the internet and the continuous expansion of rural internet penetration, an increasing number of scholars has focused on the impacts of internet use, digital technology, digital capabilities, digital literacy, etc., on green food production [25–29]. However, these studies mainly focus on single technologies or behaviors, including the adoption of straw-returning technology [30], fertilizer reduction [3], pesticide reduction [31], etc.

In summary, the literature on farmers' green production behavior is already very rich, evolving from traditional factors (such as land, labor, etc.) to digital technology. However, most studies focus on broad farmers' production behavior without considering the particularity of food production itself. Some studies involving green food production only focus on a single green production technology, lacking an overall and systematic analysis of green food production. Therefore, this paper uses data from the China Rural Revitalization Survey (CRRS) to select five production behaviors in the food production

process and analyze the overall impact of digital literacy on green food production. The potential marginal contributions of this paper lie in two aspects: First, considering the particularity of food production, it analyzes the dilemmas and paths of green transformation in food production from the characteristics of double-positive externalities and a high mechanization rate throughout the entire process, providing new ideas for studying the green transformation of agricultural production. Second, it explores the impact of digital literacy on green food production from two directions—transformation motivation and transformation path—enriching relevant mechanism research and supplementing and expanding the existing literature.

2. Theoretical Analysis and Research Hypotheses

According to the definition of the United Nations Educational, Scientific and Cultural Organization (UNESCO), digital literacy refers to the ability to safely and appropriately acquire, manage, understand, integrate, communicate, evaluate, and create information through digital technologies for the purpose of employment, decent work, and entrepreneurship [32]. Therefore, farmers with high digital literacy can obtain more information on green food production, effectively alleviate the negative impacts caused by information asymmetry [33], and promote green food production. Hence, based on the dilemmas of green food production, this paper constructs a "Digital Literacy Transformation, Motivation Transformation Path" theoretical analytical framework to systematically explore the impact of digital literacy on farmers' green food production.

Different from other crops, grain production has special transformation difficulties and realization paths based on its own characteristics. On the one hand, green grain production has double-positive externalities, not only undertaking the strategic function of ensuring national food security but also having the ecological function of protecting the environment and reducing carbon and fixing carbon. However, with the rise in the prices of agricultural materials, land, labor, and other factors, the income from grain cultivation has gradually declined and even suffered four consecutive years of losses from 2016 to 2019 [34]. At the same time, although consumers are increasingly preferring green organic food, for ordinary grain farmers, adopting green production technologies, such as precise pesticide application, efficient fertilization, water-saving irrigation, and straw returning, only reduces chemical input and cannot meet the requirements of green organic food to obtain "premium" income but, additionally, needs to bear the costs and risks of technological transformation. Therefore, the government's implementation of financial subsidy support is considered an effective method [35,36]. In 2015, China provided subsidies of CNY 450 (approximately USD 60) per hectare for the application of comprehensive integrated technologies, such as additional organic fertilizers, and CNY 225 (approximately USD 30) per hectare for planting green manure and inoculating rhizobia. However, with hundreds of millions of smallholder farmers in China, the huge and scattered audience for subsidies has led to enormous supervision and assessment costs, which easily induces speculative behaviors, such as "subsidy fraud" [37]. Meanwhile, relevant studies have shown that non-market-oriented and universal behavioral incentive methods, such as government subsidies, can only play a short-term role and cannot fundamentally motivate farmers' pro-environmental behaviors [38]. Therefore, a key approach to overcoming the challenges of green transformation of grain farmers is to enhance the endogenous driving force of farmers' green transformation. On the other hand, grain production has mechanization adaptability. With the expansion of the scale of non-agricultural transfer of rural labor, the rigid constraints of agricultural labor continue to increase, the opportunity cost of agricultural production continues to rise, and the degree of agricultural mechanization gradually increases. The tilling, planting, managing, and harvesting links in the grain production

process are relatively "standardized" and more suitable for mechanical operations. By 2022, the comprehensive mechanization rates of cultivation, planting, and harvesting of the three major grain crops of wheat, rice, and corn have reached 97.55%, 86.86%, and 90.60%, respectively, far higher than the national comprehensive mechanization rate of crop cultivation, planting, and harvesting. Agricultural green mechanization technology is a set of modern technologies that combines agricultural machinery and green technology, aiming at increasing production, reducing input, and environmental pollution, including mechanized deep fertilization, straw returning, water-saving irrigation, and no-till sowing [39]. Therefore, promoting the application of agricultural green mechanization technology in the process of grain production is the key path to promote green grain production.

The theory of planned behavior posits that behavioral attitude, subjective norm, and perceived behavioral control are the three main variables determining intentions and behaviors [40]. Enhancing cognition to change behavioral attitudes helps individuals transform their intentions into actions [41]. The blockage of grain farmers' adoption of green technologies under the traditional production model stems from the cognitive limitations of the fuzziness of technical benefits and the invisibility of ecological benefits [42]. In the process of collaborative transformation and development of digitization and greening, farmers' digital literacy is gradually improved. Grain farmers understand more diverse and comprehensive production information through the Internet, social media, etc., deconstruct and reshape traditional cognition, and give birth to a scientific green grain production concept system [43]. This cognitive iteration effectively activates the internal driving force of grain farmers' green production transformation. First, digital literacy helps to make the cost–benefit of green production explicit and enhance the benefit cognition of grain farmers. In traditional agricultural production, farmers' production decisions mostly rely on past accumulated experience. When facing green production technologies, due to the lack of a basis for judging the risks and benefits of new technologies, farmers tend to repeat the previous production model based on risk aversion psychology and refuse to adopt new technologies that may have risks. For example, when deciding whether to adopt precise pesticide application technology, farmers are cautious about it because they cannot predict whether the technology can reduce pesticide use while ensuring crop yield based on existing experience. However, with the further popularization and development of the Internet, the channels for farmers to obtain information continue to expand. Through short-video platforms, self-media live broadcasts, and other ways, farmers can more intuitively and conveniently understand the green production mode so that they can accurately perceive the input–output ratio of green technologies, eliminate the risk concerns caused by information asymmetry, and enhance the benefit cognition of green production. Second, digital literacy helps to visualize the ecological value of green production and improve farmers' ecological cognition [44]. Agricultural pollution warning cases pushed by social media, such as a lake pollution incident caused by excessive fertilization in a certain place, and a large amount of information about food safety on the Internet can effectively induce farmers' emotional resonance and environmental crisis awareness and stimulate farmers' sense of responsibility [45]. Therefore, digital literacy can systematically enhance the driving force of green production transformation and promote farmers' green production by reconstructing farmers' economic rational evaluation system and ecological value cognitive framework.

As the core human capital for agricultural digital transformation, digital literacy helps farmers reduce information asymmetry [46], lower information search costs, and promote the penetration and diffusion of green mechanized technologies in the small-scale peasant production system. First, digital literacy can help farmers better understand green mechanized technologies. Through agricultural big data platforms and intelligent terminals,

farmers can obtain principle explanations, operation cases, and ecological benefit analyses of green mechanized technologies in real time, such as by watching and learning about the soil protection mechanisms of straw-returning machines and no-till seeders. Meanwhile, through model cases from modern agricultural demonstration bases disseminated on short-video platforms, farmers can visually appreciate how green mechanized technologies deliver dual benefits: soil amelioration and yield increases. Second, digital literacy can help farmers more conveniently obtain green mechanized technologies [47]. In the context of the digital landing of the agricultural machinery purchase subsidy policy, farmers can accurately match green models suitable for local farming conditions through the "Smart Agricultural Machinery" APP and complete subsidy application, model comparison, and supplier docking online. For instance, smart agriculture zones established by select e-commerce platforms consolidate access to green equipment, such as biomass harvesters and electric precision seeders, through rental and purchase channels. By leveraging live-stream demonstrations and VR immersive experiences, farmers can transcend geographical barriers to acquire cross-regional advanced technological resources. Finally, digital literacy can help farmers better use green mechanized technologies. The intelligent monitoring system based on the Internet of Things technology can guide farmers to accurately adjust the operation parameters of agricultural machinery, such as automatically correcting the tilling depth of the subsoiler according to the Beidou navigation data and optimizing the delivery ratio of the variable fertilizer applicator with the help of soil sensors in real time. By participating in digital agricultural technology training courses, farmers can systematically master core skills, such as battery maintenance of electric plant protection machines and path planning of driverless tractors. Therefore, digital literacy can effectively promote the green production of grain farmers by promoting the application of green mechanized technologies (Figure 1).

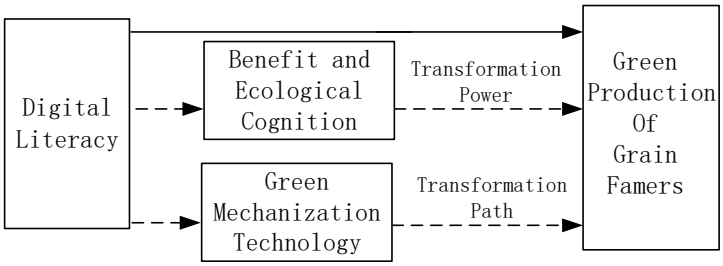


Figure 1. Theoretical framework diagram.

Based on this, Hypotheses 1, 2, and 3 are proposed.

Hypothesis 1. *Digital literacy can promote the green production of grain farmers.*

Hypothesis 2. *Digital literacy promotes the green production of grain farmers by reconstructing benefit cognition and ecological cognition.*

Hypothesis 3. *Digital literacy promotes the green production of grain farmers by promoting the application of green mechanized technologies.*

3. Data and Empirical Methods

3.1. Data Source

The data used in this study comes from the 2020 China Rural Revitalization Survey conducted by the Institute of Rural Development, Chinese Academy of Social Sciences, referred to as the "China Rural Revitalization Comprehensive Survey" (CRRS) national survey data. The data comprehensively considers factors such as socioeconomic develop-

ment level, geographical location, and agricultural–rural development status. Following the principle of random stratified sampling, it first selects 10 provinces from the eastern, central, western, and northeastern regions at a ratio of 1/3 of the number of provinces in each division. Secondly, all counties (cities, districts) in each province are divided into 5 groups based on per capita GDP levels. Considering the uniform geographical distribution of counties, 1 county is randomly selected from each group, meaning 5 counties (cities, districts) are drawn from each province. Thirdly, according to high, medium, and low economic development levels, 3 townships (towns) are randomly selected in each county (city, district), and each township (town) selects 1 administrative village with better economic development and 1 with poorer economic development based on local economic conditions. Finally, investigators (professors and trained postgraduate students in related disciplines) use equidistant sampling to randomly select 12–14 rural households from the roster of each administrative village, conducting surveys on agricultural production, rural development, household income, etc. The data covers multiple aspects, such as agricultural production, rural development, farmers' lives, and social welfare, ensuring sufficient representativeness. Since this paper studies the impact of digital literacy on farmers' green production, combined with the design of the questionnaire and the actual needs of the research, farmers actually engaged in the cultivation of corn, wheat, and rice were selected, samples with missing data were excluded, and, finally, 1811 farmer samples were obtained.

3.2. Variable Setting

3.2.1. Dependent Variable: Green Production Behavior

According to the definition of the United Nations Environment Programme, green production behavior refers to a production mode that can not only ensure and increase agricultural productivity and profitability but also reduce rural environmental pollution and improve resource utilization efficiency [48]. Agricultural materials, such as fertilizers and pesticides, are important sources of carbon emissions in agricultural production [49]. Reducing the use of fertilizers and pesticides can not only reduce carbon emissions but also protect soil health and ensure the sustainability of grain production. Green technologies, such as straw returning and water-saving irrigation, can reduce resource consumption and effectively play the role of soil's carbon sequestration capacity. The random disposal of pesticide packaging will cause the residual pesticides in the packaging to penetrate into the soil, damage the soil structure. Moreover, most pesticide packaging is plastic products, which take a long time to naturally degrade, and long-term accumulation will lead to "white pollution". Therefore, the recycling of pesticide packaging is also important. Based on the analysis above and combined with the survey data, five aspects of whether to carry out fertilizer reduction, pesticide reduction, water-saving irrigation, straw recycling, and reasonable disposal of pesticide packaging are selected to define farmers' green production behavior. If farmers adopt a certain behavior in the process of planting corn, wheat, and rice, it is assigned a value of 1, and if not, it is assigned a value of 0. Then, the values of the 5 variables are summed to characterize farmers' green behavior.

3.2.2. Explanatory Variable: Farmers' Digital Literacy

In the context of the information and digital era, digital literacy has become an important indicator to measure an individual's comprehensive quality and social adaptability and is an important part of human capital. Referring to the Global Digital Literacy Framework released by the United Nations Educational, Scientific and Cultural Organization and existing research [32], based on the content of the CRRS questionnaire, this paper measures farmers' digital literacy from four dimensions and eleven measurement items, including

digital device access, digital information acquisition, digital communication and social interaction, and digital production and operation, as shown in Table 1.

Table 1. Digital literacy measurement index system.

Dimension	Measurement Items	Assignment
Digital Device Access	Do you use a 4G/5G mobile phone?	Yes = 1, No = 0
	Does your family have a computer?	Yes = 1, No = 0
	How is your home network condition?	Very good or acceptable = 1, poor = 0
	Have you received training in computer or mobile phone Internet access?	Yes = 1, No = 0
Digital Information Acquisition	Do you think the information obtained through the network can meet the daily needs of production and life?	Yes = 1, No = 0
	If there are daily needs, can you obtain relevant information at any time through your mobile phone or network by yourself?	Yes = 1, No = 0
Digital Communication and Social Interaction	Do you use smart phone devices for chat and social activities, such as WeChat and Weibo?	Yes = 1, No = 0
	Have you ever communicated with the village on important public affairs through WeChat groups?	Yes = 1, No = 0
Digital Production and Operation	Have you received e-commerce training and guidance services?	Yes = 1, No = 0
	Does your family operate products traded through the network?	Yes = 1, No = 0
	Do you purchase seeds, fertilizers, and other agricultural materials through online payment?	Yes = 1, No = 0

3.2.3. Control Variables

In order to control the influence of other confounding factors on farmers' green production as much as possible, this paper selects control variables from the aspects of household head characteristics, family characteristics, institutional environment characteristics, and village characteristics. Specifically, in terms of household head characteristics, the gender, age, political affiliation, and education level of the household head are selected; in terms of family characteristics, the management model and the number of agricultural laborers are selected; in terms of institutional environment characteristics, whether there is a recycling point for waste agricultural films or pesticide packaging is selected; and in terms of village characteristics, the location, logistics, and terrain conditions of the village are selected.

3.2.4. Mechanism Variables

In order to test the internal mechanism of digital literacy affecting the green production behavior of grain farmers, this paper selects "whether it is considered that straw returning as fertilizer reduces the cost of farming" to measure farmers' benefit cognition; selects "whether it is considered that pesticide packaging will pollute the environment" to measure farmers' ecological cognition; and selects the proportion of mechanical pesticide application to represent the application of green mechanization technology (Table 2).

Table 2. Variable definitions and descriptive statistics.

	Variable	Definition	Mean	Standard Deviation
Dependent Variable	Green Production Behavior	Number of kinds of green production behaviors of farmers	2.294	0.967
Independent Variable	Digital Literacy Score	Entropy Method	0.190	0.160
		Principal Component Analysis	0.892	1.328
		Principal Component Analysis	0.405	0.206
Control Variables	Age	Age	55.252	10.607
	Gender	Male = 0, Female = 1	0.045	0.207
	Political Identity	Communist Party member = 1, Others = 0	0.210	0.408
	Education	1 = No schooling; 2 = Primary school; 3 = Junior high school; 4 = Senior high school; 5 = Technical secondary school; 6 = Vocational high school and technical school; 7 = Junior college; 8 = Undergraduate and above	2.724	0.979
	Management Mode	Whether joined a cooperative or applied to be a family farm: Yes = 1, No = 0	0.261	0.439
	Agricultural Laborers	Number of people engaged in agricultural labor	2.041	0.852
	Pesticide Packaging Recycling Station	Whether there is a recycling point for waste agricultural films or pesticide packaging: Yes = 1, No = 0	0.397	0.490
	Village Logistics Conditions	Whether express delivery can reach the household or there is a pickup point: Yes = 1, No = 0	0.773	0.420
	Village Terrain	1 = Plain; 2 = Hilly area; 3 = Mountainous area; 4 = Semi-mountainous area	1.831	0.890
	Village Location	Distance from the county seat	23.574	16.107
Mediating Variables	Benefit Cognition	Whether it is considered that returning straw to the field as fertilizer reduces farming costs: Yes = 1, No = 0	0.533	0.500
	Ecological Cognition	Whether it is considered that pesticide packaging will pollute the environment: Yes = 1, No = 0	0.734	0.442
	Green Mechanization Technology	Proportion of mechanical pesticide application	0.278	0.444

3.3. Empirical Model

3.3.1. The Impact of Digital Literacy on Farmers' Green Production Behavior

Since the numbers of green production behaviors of grain farmers are 0, 1, 2, 3, 4, and 5, with obvious sequential relations, the Order Probit model was selected to estimate the impact of digital literacy on the green production behavior of grain farmers. The model is designed as follows:

$$Y_i = \alpha_0 + \alpha_1 Digital_literacy_i + \alpha_2 X_i + \mu_i \quad (1)$$

Y_i represents the green production behavior of farmer i; $Digital_literacy_i$ is the digital literacy of farmer i; X_i represents a series of control variables, including farmer individual characteristics, family characteristics, management characteristics, village characteristics, etc.; μ_i is the random disturbance term; α_0 is the constant term; and α_1, α_2 are the parameters to be estimated. Assuming that μ follows an N(0,1) distribution, the Order Probit model can be expressed as follows:

$$\begin{aligned}
P(Y_i = 0|x) &= P(Y^* \le r_0|x) \\
&= \varphi(r_0 - \alpha_1 Digital_literacy_i - \alpha_2 X_i) \\
P(Y_i = 1|x) &= P(r_0 < Y^* \le r_1|x) \\
&= \varphi(r_1 - \alpha_1 Digital_literacy_i - \alpha_2 X_i) - \\
&\quad \varphi(r_0 - \alpha_1 Digital_literacy_i - \alpha_2 X_i) \\
&\cdots\cdots \\
P(Y_i = 5|x) &= P(r_4 \le Y^*|x) \\
&= 1 - \varphi(r_4 - \alpha_1 Digital_literacy_i - \alpha_2 X_i)
\end{aligned} \quad (2)$$

Among them, $r_0 < r_1 < r_2 < r_3 < r_4$ are the parameters to be estimated; the values of Y_i are 0, 1, 2, 3, 4, 5, respectively, indicating that farmers "do not adopt" to "adopt 5 kinds of green production behaviors", and φ is the cumulative density function of the standard normal distribution. By constructing the green production function of each farmer, the maximum likelihood method is used to estimate the model parameters.

3.3.2. The Action Mechanism of Digital Literacy Affecting Farmers' Green Production

In order to further explore the specific influence mechanism of digital literacy on farmers' green production behavior, combined with Formula (1), this paper constructs a regression model of mechanism variable M_i and core explanatory variable $Digital_literacy_i$ to test the impact of digital literacy on improving farmers' benefit cognition and green cognition and promoting the application of green mechanization technology. It also constructs a regression model of farmers' green production behavior Y_i and all mechanism variables M_i to investigate the action mechanism of digital literacy on farmers' green production. The model is set as follows:

$$M_i = \beta_0 + \beta_1 Digital_literacy_i + \beta_2 X_i + \varepsilon_{1i} \quad (3)$$

$$Y_i = \gamma_0 + \gamma_1 Digital_literacy_i + \gamma_2 M_i + \gamma_3 X_i + \varepsilon_{2i} \quad (4)$$

Among them, $\beta_1, \beta_2, \gamma_1, \gamma_2, \gamma_3$ are the parameters to be obtained, and ε_i is the error term.

4. Analysis of Empirical Results

4.1. Benchmark Regression Results

In order to empirically test the impact of digital literacy on the green production of grain farmers, regression analysis is carried out using the Order Probit model based on Formula (1), and robust standard errors are used. The regression results are shown in Table 3. Specifically, control variables are gradually added from the simplest estimation to minimize the estimation bias caused by omitted variable problems. Column (1) does not control any possible related factors, column (2) controls household head characteristic variables, column (3) adds control of family management characteristic variables, column (4) adds village characteristic variables, and column (5) adds province fixed effects. This paper takes the results of column (5) as the benchmark regression results for analysis. The results show that digital literacy significantly promotes farmers' green production at the 1% statistical level.

Table 3. The benchmark regression results.

	(1)	(2)	(3)	(4)	(5)
Digital Literacy	0.921 ***	0.846 ***	0.809 ***	0.662 ***	0.584 ***
	(0.158)	(0.177)	(0.182)	(0.190)	(0.194)
Age		0.002	0.002	0.000	−0.006 **
		(0.003)	(0.003)	(0.003)	(0.003)
Gender		0.042	0.039	0.055	−0.008
		(0.124)	(0.125)	(0.126)	(0.131)
Political Identity		0.026	0.013	0.060	0.037
		(0.064)	(0.064)	(0.066)	(0.067)
Education		0.059 **	0.058 **	0.014	0.000
		(0.029)	(0.029)	(0.030)	(0.031)
Number of Agricultural Laborers			−0.030	−0.021	−0.008
			(0.031)	(0.032)	(0.033)
Management Mode			0.115 **	0.113 *	0.113 *
			(0.058)	(0.059)	(0.062)
Pesticide Packaging Recycling Station				0.171 ***	0.256 ***
				(0.054)	(0.058)
Terrain				−0.270 ***	−0.293 ***
				(0.034)	(0.040)
Logistics Conditions				−0.018	−0.028
				(0.060)	(0.064)
Distance from County Seat				−0.003	−0.002
				(0.002)	(0.002)
Province Control Effect	No	No	No	No	Yes
N	1811	1804	1804	1745	1745
Pseudo R^2	0.007	0.009	0.010	0.036	0.055

Note: ***, **, and * represent the significance levels of 1%, 5%, and 10%, respectively, and the numbers in parentheses are the standard errors of the coefficients, the same as the following tables.

From the perspective of control variables, based on household head characteristic variables, the age of the household head has a negative impact on green production, which may be because the older the farmer, the stronger the behavior habits, which is

not conducive to the green production transformation. From the perspective of family management characteristics, participation in cooperatives or becoming a family farm has a significant positive impact on the green production behavior of grain farmers. After participating in cooperatives or becoming family farms, farmers can obtain more production guidance services, etc., providing various supports for farmers' green production, thereby promoting farmers to implement green production behaviors. From the perspective of village characteristics, the presence of a pesticide packaging recycling point in the village has a significant positive impact on the green production of grain farmers. This may be because the establishment of a pesticide packaging recycling point reflects to a certain extent the resources invested by the local government in green production. The more the local government attaches importance, the more conducive it is to green production. The village terrain has a significant negative impact on the green production of grain farmers. The steeper the terrain, the more backward the corresponding farmland water conservancy and transportation infrastructure, which is not conducive to the use of green mechanized technologies, thereby reducing the probability of grain farmers implementing green production behaviors.

In the Order Probit model, the benchmark regression analysis mainly reveals the statistical significance and direction of action (the sign of the β coefficient) of the explanatory variables and cannot quantitatively analyze the impact of the core explanatory variables on the green production of grain farmers. Therefore, this paper further calculates the marginal effect of grain farmers' digital literacy on green production behavior, intuitively presenting the magnitude of the change in the decision probability caused by the unit change of the explanatory variable. Column (1) of Table 4 calculates the marginal effect of digital literacy on the green production behavior of grain farmers. Specifically, a one-unit increase in digital literacy reduces the probability of the zero-adoption state by 3.4 percentage points and the single-adoption probability by 17.5 percentage points, while the probability of multi-combination adoption shows a gradient response, and the probabilities of adopting two to five kinds increase by 5.7, 10.8, 3.6, and 0.7 percentage points, respectively.

Table 4. Marginal effects.

Green Production Behavior	Benchmark Regression (1)	CMP (2)
Not Adopted	−0.034 ***	−0.482 *
	(0.011)	(0.279)
Adopted One Kind	−0.175 ***	−0.833 ***
	(0.057)	(0.090)
Adopted Two Kinds	0.057 ***	0.278 ***
	(0.020)	(0.066)
Adopted Three Kinds	0.108 ***	0.517 ***
	(0.035)	(0.065)
Adopted Four Kinds	0.036 ***	0.310 ***
	(0.012)	(0.092)
Adopted Five Kinds	0.007 **	0.211
	(0.003)	(0.160)

4.2. Discussion on Endogeneity Issues

With the continuous improvement of agricultural digitization, when farmers carry out green production, they may contact and use more digital devices, thereby improving their own digital literacy. Therefore, the benchmark results of this paper may have estimation bias caused by the endogeneity problem of mutual causation. In order to solve this problem, this study uses the Conditional Mixed Processes (CMP) method to correct the estimation

results [50]. In this paper, the main equation and the auxiliary equation are the Order Probit model and the OLS model, respectively. The dependent variable of the main equation is the number of green production behaviors of farmers, and the independent variables are digital literacy, control variables, and province fixed effects; the dependent variable of the auxiliary equation is digital literacy, and the independent variables are instrumental variables, control variables, and province fixed effects.

According to the conditions that the instrumental variable is highly correlated with the endogenous explanatory variable but not correlated with the error term, this paper uses the Internet access rate of the interviewed farmers' village as the instrumental variable of farmers' digital literacy. In terms of correlation, the Internet access rate of the village reflects the level of digital infrastructure in the village to a certain extent, and its public good attribute has a spatial spillover effect, which will have a positive impact on the understanding and use of network digital technology by the residents in the village. In terms of exogeneity, the Internet access rate is mainly used for information transmission functions, has no direct technical complementarity with agricultural production technologies, and has no direct impact on farmers' green production behavior.

As shown in the auxiliary equations of columns (1) and (2) of Table 5, regardless of whether province fixed effects are controlled, the regression coefficient of the broadband access rate on digital literacy is significantly positive at the 1% level, indicating that the instrumental variable has a strong correlation with the endogenous variable. At the same time, it can be seen from Table 5 that the CMP statistic atanhrho_12 is significant, at least at the 5% level, indicating that digital literacy is an endogenous variable, and it is also reasonable to use the CMP model to handle endogeneity. Compared with the estimation results of the Order Probit model in Table 3, the estimation coefficient of digital literacy in the CMP model is still significantly positive, but its absolute value has increased significantly. This indicates that due to the potential endogeneity problem, the Order Probit model underestimated the promoting effect of digital literacy on farmers' green production. In addition, the marginal effects in column (2) of Table 4 also confirm this conclusion. After controlling the endogeneity problem, a one-unit increase in digital literacy reduces the probability that farmers do not adopt any green production behavior by 48.2% and the probability of adopting one kind by 83.3%, while the probabilities of adopting two to five kinds increase by 27.8%, 51.7%, 31.0%, and 21.1% respectively, all higher than the marginal effects in the benchmark regression results.

Table 5. CMP estimation.

Variables	(1)		(2)	
	Auxiliary Equation	**Main Equation**	**Auxiliary Equation**	**Main Equation**
Digital Literacy		5.619 ***		4.057 ***
		−0.557		−1.161
Internet Access Rate	0.045 ***		0.046 ***	
	−0.009		−0.011	
Control Variables	Yes	Yes	Yes	Yes
Province Fixed Effects	No	No	Yes	Yes
N	1745	1745	1745	1745
atanhrho_12	−0.905 ***	—	−0.534 **	—

4.3. Robustness Test

4.3.1. Replacing the Measurement Method of the Key Explanatory Variable of Farmers' Digital Literacy

In the benchmark regression, this paper uses the entropy method to calculate the weight of the farmers' digital literacy index. However, in the existing literature, the principal component analysis method and the equal weight method are also widely used in constructing comprehensive indicators. Therefore, this paper further uses the principal component analysis method and the equal weight method to recalculate the level of farmers' digital literacy and respectively uses the Order Probit model and the CMP method to carry out empirical tests. The results are shown in Table 6. The regression results show that digital literacy is still significant and the coefficient is positive. It should be noted that the coefficient in the first column of Table 6 is smaller than that in the benchmark regression because after measuring digital literacy by the principal component analysis method, the value range of the independent variable increases.

Table 6. Robustness test: using alternative measurement approaches for explanatory variables.

Variables	(1) Principal Component Analysis		(2) Equal Weight Method	
	Order Probit	**CMP**	**Order Probit**	**CMP**
Digital Literacy	0.082 ***	0.503 ***	0.509 ***	3.205 ***
	(0.025)	(0.146)	(0.163)	(0.907)
Control Variables	Yes	Yes	Yes	Yes
Province Fixed Effects	Yes	Yes	Yes	Yes
N	1614	1614	1614	1614
R^2	0.052	—	0.052	
atanhrho_12	—	−0.521 **		−0.514 ***

4.3.2. Re-Estimating by Replacing the Estimation Model

The ordered logit model is introduced to analyze the impact of digital literacy on farmers' green production behavior, and the farmers' digital literacy calculated by different measurement methods is respectively substituted. The results are shown in Table 7. The regression results show that the impact of digital literacy on farmers' green production behavior is still significant and the coefficient is positive.

Table 7. Robustness test: employing alternative model specifications.

Variables	Ordered Logit		
	Entropy Method	**Principal Component Analysis**	**Equal Weight Method**
Digital Literacy	1.093 ***	0.148 ***	0.909 ***
	(0.346)	(0.045)	(0.286)
Control Variables	Yes	Yes	Yes
Province Fixed Effects	Yes	Yes	Yes
N	1745	1614	1614
R^2	0.056	0.052	0.052

The results of the robustness test consistently show that digital literacy has a significant positive impact on farmers' green production behavior, that is, the benchmark regression results have strong robustness.

4.4. Mechanism Test

In order to reveal the mechanism through which digital literacy affects the green production of grain farmers, this paper starts with the motivation and path of green production transformation and carries out an analysis of the internal mechanism of digital literacy promoting green production. The results are shown in Table 8.

Table 8. Mechanism test.

Variables	Benefit Cognition (1)	Ecological Cognition (2)	Mechanical Pesticide Application (3)	Green Production (4)	Green Production (5)	Green Production (6)
Digital Literacy	0.512 **	0.634 **	0.715 **	0.538 ***	0.555 ***	0.556 ***
	(0.232)	(0.258)	(0.292)	(0.196)	(0.194)	(0.195)
Benefit Cognition				0.279 ***		
				(0.056)		
Ecological Cognition					0.155 ***	
					(0.059)	
Mechanical Pesticide Application						0.178 **
						(0.070)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Province Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
N	1745	1745	1745	1745	1745	1745
R^2/Pseudo R^2	0.103	0.036	0.363	0.061	0.057	0.057

4.4.1. Reconstructing Benefit and Ecological Cognition and Enhancing the Driving Force of Green Transformation

As shown in column (1) of Table 8, the improvement of farmers' digital literacy can enhance farmers' benefit cognition, so that in the process of agricultural production, they gradually realize that green production methods, such as water-saving irrigation and straw recycling, not only do not bring income losses but also can reduce production costs, protect soil quality, and increase income. As shown in column (2) of Table 8, the improvement of farmers' digital literacy can enhance farmers' ecological cognition, and more and more pay attention to environmental protection and greenness in agricultural production. Columns (4) and (5) of Table 8 correspond to the impact of benefit cognition and ecological cognition on farmers' green production behavior. It can be found that the improvement of cognition significantly promotes farmers' green production behavior. This indicates that digital literacy can promote farmers' green production behavior by reconstructing benefit and ecological cognition. Thus, Hypothesis 2 is verified.

4.4.2. Promoting the Application of Green Mechanization Technology and Broadening the Path of Green Transformation

It can be seen from the results of column (3) of Table 8 that digital literacy is conducive to promoting the application of green mechanization technology, which indicates that digital literacy can provide help for farmers to learn, obtain, and use green mechanization technology. Column (6) of Table 8 corresponds to the impact of the application of green mechanization technology on farmers' green production behavior. It can be found that the application of green mechanization technology significantly promotes farmers' green production behavior, indicating that the intermediate mechanism of the application of green mechanization technology is realized. Thus, Hypothesis 3 is verified.

4.5. Heterogeneity Analysis

In reality, there are great differences among farmers, and the impact of digital literacy on different types of farmers is also different. Therefore, this paper conducts group regres-

sion from the two aspects of farmers' operation scale and economic capacity to analyze the heterogeneous impact of digital literacy on the green production behavior of grain farmers.

The operation scale may have significant differences in the impact on farmers' green production. From the perspective of farmers' cognition, small-scale farmers have strong risk aversion psychology and weak green technology cognition. With the expansion of the operation scale, farmers' technical cognition level and factor input preference will change, and they are more likely to accept the green production concept. In terms of cost-benefit, small-scale farmers have limited funds and find it difficult to share the high marginal cost of green technology, while large-scale farmers have higher mechanized operation efficiency, less loss, and lower cost, which is conducive to promoting the application of green mechanized technologies, such as mechanized straw returning and water-saving irrigation. In terms of production conditions, small-scale farmers are restricted by land fragmentation and backward infrastructure, while large-scale farmers with contiguous land, specialized equipment, and risk resistance are more likely to achieve efficient adaptation of green technologies. Therefore, according to the mean value (2.07 hectares) of the sample farmers' operation scale, the sample farmers above the mean are regarded as large farmers, and the sample farmers below the mean are regarded as small farmers. Thus, group regression is carried out on the farmers, and the results are shown in columns (1) and (2) of Table 9. The impact coefficient of digital literacy on the green production behavior of large-scale farmers is significantly positive at the 1% level. For small-scale farmers, the impact coefficient is only significantly positive at the 5% level, and the coefficient itself is also smaller than that of large-scale farmers. It can be seen that the larger the operation scale, the more significant the promoting effect of digital literacy on farmers' green production behavior.

Table 9. Heterogeneity test.

Variables	Operation Scale		Farmer Income	
	Small Farmers (1)	Large Farmers (2)	Low Income (5)	High Income (6)
Digital Literacy	0.470 **	1.143 ***	0.502 *	0.760 **
	(0.231)	(0.426)	(0.269)	(0.307)
Control Variables	Yes	Yes	Yes	Yes
Province Fixed Effects	Yes	Yes	Yes	Yes
N	1409	336	1204	541
R^2/Pseudo R^2	0.055	0.099	0.049	0.086

Economic capacity determines the differences in resource endowments, such as land and agricultural machinery owned by farmers, which will affect farmers' production and operation decisions. Specifically, high-income farmers have stronger economic capacity and more productive resources. Under the empowerment of digitization, the optimization of resource allocation efficiency is more obvious [32], which is more conducive to the adoption of green production technologies. At the same time, compared with low-income farmers, high-income farmers generally have more social capital, which will bring them more opportunities for communication and learning, which is conducive to their understanding of agricultural green production. The difference in economic capacity will also lead to digital inequality. Low-income groups tend to use digital technologies for leisure and entertainment, while high-income groups tend to use digital technologies for productive activities. Therefore, this paper refers to the existing literature [51] and uses the household income of farmers as a proxy variable to describe the differences in the economic capacity

of farmer households. Specifically, according to the average value (CNY 70,026) of the household income of farmers, the farmers above the mean are regarded as high-income farmers, and the farmers below the mean are regarded as low-income farmers. Thus, group regression is carried out on the farmers, and the results are shown in columns (3) and (4) of Table 9. The impact coefficient of digital literacy on the green production behavior of low-income farmers is significantly positive at the 10% level, but for high-income group farmers, the impact coefficient is significantly positive at the 5% level, and the coefficient itself is also larger than that of the low-income group. Therefore, the higher the income, the more significant the promoting effect of digital literacy on farmers' green production behavior.

5. Discussion

This study is based on the 2020 CRRS survey data, selecting 1811 farmer samples engaged in grain cultivation. Using the ordered Probit model and mediation effect model, it systematically analyzes the green transformation of grain production, whether digital literacy can promote farmers' green production, how it works, and the heterogeneous performance of these impacts among different farmers.

This study finds that digital literacy can significantly promote farmers' green production behaviors. After considering endogeneity issues and conducting a series of robustness tests, the conclusion remains valid, which is consistent with other studies. For example, Wei Jiajing et al. found that among pear growers, the use of digital technologies can promote farmers' adoption of green prevention and control technologies [43]. Hong Mingyong et al. macroscopically discovered that the digital economy has a positive role in promoting agricultural green development [52].

Different from existing research, this paper supplements the literature by analyzing the promoting effect of digital literacy from the perspectives of green transformation motivation and the transformation path. In terms of transformation motivation, enhancing farmers' benefit cognition and green cognition can effectively promote green production, which is consistent with the findings of Michael et al. in Indonesia [53]. Meanwhile, considering the particularity of grain cultivation, this paper explores the transformation path through green agricultural mechanization technologies. Due to the high mechanization rate in the entire grain production process, digital literacy can improve farmers' green production by promoting the application of green mechanization technologies. Additionally, this study finds that flatter terrain is more conducive to farmers' green production, possibly because flatter terrain facilitates mechanized operations.

In the heterogeneity analysis, we expand the research into two aspects: farmers' operation scale and economic capacity. The conclusions are both consistent with and different from existing studies. We find that larger operation scales are more conducive to green production, which aligns with the research of Dingde [54], Xiong Feixue, etc., but differs from Du Yu et al.'s [25] findings—this may be because Du Yu et al. focused on fruit growers rather than grain producers. In terms of economic capacity, we find that the stronger the economic capacity, the more likely farmers are to engage in green production.

Although some scholars have analyzed the role of information technology and digital literacy in farmers' green production, few have considered the particularity of grain production and the motivation and path of farmers' green transformation. Therefore, starting from the dilemmas of green grain production, this paper finds that digital literacy can enhance farmers' green transformation motivation by reconstructing their benefit and green cognition, and broaden their transformation path by promoting the application of green mechanization technologies.

6. Implications and Prospects

Based on the research findings above, this study proposes the following key policy implications to effectively promote sustainable agricultural transformation among smallholders globally, especially in developing countries: First, systemic investment is needed to bridge the rural digital divide. This includes strengthening digital infrastructure and conducting targeted digital skills training to enhance smallholders' capacity to access and apply market information, precision agriculture tools, and financial services. Second, effective economic incentive mechanisms should be designed to overcome barriers to green transformation. Examples include implementing performance-based subsidies linked to specific sustainable practices (e.g., conservation tillage, precision pesticide application) and developing complementary insurance products to manage potential yield reduction risks during the initial transition phase, thereby tangibly strengthening adoption incentives. Third, accelerating the promotion of green mechanization technologies suitable for smallholders is crucial. This requires building a comprehensive support system through policy incentives (e.g., optimizing agricultural machinery subsidies), specialized R&D to address key technological bottlenecks (e.g., small-scale smart equipment), establishing grassroots technical service networks and operator training/certification systems, and innovating socialized service models (e.g., machinery rental cooperatives). This will enable smallholders to conveniently and affordably access advanced green production machinery and services. Collectively, these measures aim to lower the barriers to transition, manage risks, and enhance capabilities. They hold universal reference value for promoting the sustainability and inclusive development of global food systems.

Simultaneously, this study has several limitations: First, concerning the indicator system for digital literacy, the survey questionnaire design itself posed issues. The scale contained too many binary (yes/no) items, which hindered accurate measurement of farmers' true digital literacy. This aspect requires significant improvement in future surveys. Second, the data used constitutes a cross-sectional dataset collected in 2020. As this data has already been utilized in several other papers [55–59], it lacks novelty. Longitudinal survey data over multiple years is needed to better examine the role of digital literacy. Additionally, although this study analyzes green grain production holistically, it only selected five specific green production practices. This is insufficient to cover the entire grain production process.

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